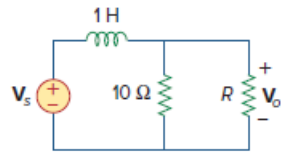


Problem

A periodic signal given by $v_s(t) = 10 \text{ V}$ for $0 < t < 1$ and 0 V for $1 < t < 2$ is applied to the high-pass filter in Fig. 17.87. Determine the value of R such that the output signal $v_o(t)$ has an average power of at least 70 percent of the average power of the input signal.

Figure 17.87:



Step-by-step solution

Step 1 of 1

$v_s(t)$ is the same as $f(t)$ in Figure 16.1 except that the magnitude is multiplied by 10. Hence

$$v_o(t) = 5 + \frac{20}{\pi} \sum_{k=1}^{\infty} \frac{1}{n} \sin(n\pi t), \quad n = 2k - 1$$

$$T = 2, \quad \omega_o = 2\pi/T = \pi, \quad \omega_n = n\omega_o = 2n\pi$$

$$j\omega_n L = j2n\pi; \quad Z = R \parallel 10 = 10R/(10 + R)$$

$$V_o = ZV_s/(Z + j2n\pi) = [10R/(10R + j2n\pi(10 + R))]V_s$$

$$V_o = \frac{10R \angle -\tan^{-1}\{(n\pi/5R)(10 + R)\}}{\sqrt{100R^2 + 4n^2\pi^2(10 + R)^2}} V_s$$

$$V_s = [20/(n\pi)] \angle 0^\circ$$

The source current I_s is

$$\begin{aligned} I_s &= \frac{V_s}{Z + j2n\pi} = \frac{V_s}{\frac{10R}{10 + R} + j2n\pi} = \frac{(10 + R) \frac{20}{n\pi}}{10R + j2n\pi(10 + R)} \\ &= \frac{(10 + R) \frac{20}{n\pi} \angle -\tan^{-1}\{(n\pi/5)(10 + R)\}}{\sqrt{100R^2 + 4n^2\pi^2(10 + R)^2}} \end{aligned}$$

$$p_s = V_{DC}I_{DC} + \frac{1}{2} \sum V_{sn} I_{sn} \cos(\theta_n - \phi_n)$$

For the DC case, L acts like a short-circuit.

$$I_s = \frac{5}{\frac{10R}{10 + R}} = \frac{5(10 + R)}{10R}, \quad V_s = 5 = V_o$$

$$\begin{aligned} p_s &= \frac{25(10 + R)}{10R} + \frac{1}{2} \left[\left(\frac{20}{\pi} \right)^2 \frac{(10 + R) \cos\left(\tan^{-1}\left(\frac{\pi}{5}(10 + R)\right)\right)}{\sqrt{100R^2 + 4\pi^2(10 + R)^2}} \right. \\ &\quad \left. + \left(\frac{10}{\pi} \right)^2 \frac{(10 + R)^2 \cos\left(\tan^{-1}\left(\frac{2\pi}{5}(10 + R)\right)\right)}{\sqrt{100R^2 + 16\pi^2(10 + R)^2}} + \dots \right] \end{aligned}$$

$$\begin{aligned} p_s &= \frac{V_{DC}}{R} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{V_{on}}{R} \\ &= \frac{25}{R} + \frac{1}{2} \left[\frac{100R}{100R^2 + 4\pi^2(10 + R)^2} + \frac{100R}{100R^2 + 16\pi^2(10 + R)^2} + \dots \right] \end{aligned}$$

We want $p_o = (70/100)p_s = 0.7p_s$. Due to the complexity of the terms, we consider only the DC component as an approximation. In fact the DC component has the largest share of the power for both input and output signals.

$$\frac{25}{R} = \frac{7}{10} \times \frac{25(10 + R)}{10R}$$

$$100 = 70 + 7R \quad \text{which leads to } R = 30/7 = \mathbf{4.286\Omega}$$